Supplementary Material

Simulation design and analysis

Main sweeps. The main simulations use neural-network learning and run for $T = 500$ periods. Table S1 gives the parameters varied one at a time; Table S2 gives the crossed designs. Parameters outside each sweep remain at baseline (Appendix B, §9). The reservation coefficient f_r sets the threshold $r = f_r \bar{q}_{\text{cal}}$.

Table S1: Baseline values and one-at-a-time sweeps.

Parameter	Baseline	Values varied
Matching composition, ρ	0.5	0, 0.15, 0.3, 0.5, 0.7, 0.85, 1
Turnover probability, η	0.02	0, 0.001, 0.01, 0.02, 0.03
Number of principals, N	1000	500, 1000, 1500
Reservation coefficient, f_r	0.6	0.4, 0.6, 0.9, 1.2
Matching difficulty, δ	0.5	0, 0.25, 0.5, 0.75, 1
Initial ring degree, k_G	6	4, 6, 12
Broker roster share, α_R	0.2	0.1, 0.2, 0.4
Self-search stranger pool, $n_{\text{strangers}}$	10	0, 10, 50

Table S2: Crossed designs. Each row includes all combinations of the listed values.

Parameters	First parameter	Second parameter
$\rho \times \eta$	0, 0.15, 0.5, 0.85, 1	0, 0.001, 0.01, 0.02, 0.03
$\rho \times f_r$	0, 0.5, 1	0.4, 0.6, 0.9, 1.2
$\eta \times f_r$	0, 0.001, 0.01, 0.02, 0.03	0.4, 0.6, 0.9, 1.2
$\rho \times \delta$	0, 0.15, 0.3, 0.5, 0.7, 0.85, 1	0, 0.25, 0.5, 0.75, 1

The sweeps contain 80 distinct regimes. Each uses seeds 1–20; the baseline uses seeds 1–50, giving 1,630 runs per learning model. Configurations shared by sweeps use the same runs. At $\rho = 1$, δ has no effect, so its grid coordinates share one regime.

Learning comparisons. The complete design is repeated with ridge regression. Restrictions on the ridge broker assess the contributions of observation volume and information about both parties: a fitting sample capped at the median principal sample size, one party recorded per match, and omission of the interaction term. Each restriction is compared with the unrestricted ridge broker across the 31 distinct regimes in the $\rho \times \delta$ grid, including the baseline. Each restriction uses 650 runs with the same seed allocation as the main sweeps. Principal learning rules are unchanged.

Broker services. The neural-network model compares full service, assessment-only service, and access-only service. The comparison crosses $\rho \in \{0, 0.5, 1\}$ with $\eta \in \{0, 0.01, 0.03\}$, and adds $(\rho, \eta) = (0.5, 0.001)$ and $(\rho, \eta) = (0.5, 0.02)$. Other parameters remain at baseline. Each service uses 50 seeds at the baseline and 20 elsewhere: 11 market regimes and 750 runs across all services.

Training calibration. Separate baseline calibration runs select neural-network training settings. The screen varies learning rates $\{0.003, 0.01, 0.03\}$ and recurrent update budgets $\{50, 100, 200\}$, one

learner at a time, with initial budgets twice the recurrent budget. It uses 17 configurations, 3 seeds per configuration, and 200 periods. Selection considers each learner's ranking performance, its stability across periods and seeds, and runtime. The selected learning rates are 0.003 for principals and 0.03 for the broker, with 100 initial and 50 recurrent steps for both. Confirmation uses 5 seeds, including the screening seeds.

Ridge calibration crosses the penalties $\{0.0003, 0.001, 0.003, 0.01, 0.03\}$ for principals and the broker, using 10 seeds per combination and 500 periods. Selection considers each learner's ranking performance, stability, and prediction error. The reported experiments use penalties of 0.001 for both learners. Confirmation uses 5 new seeds. Calibration seeds are excluded from the reporting runs.

Analysis. Early and late summaries cover periods 51–70 and 481–500. Betweenness, constraint, and effective size use snapshots at periods 60 and 500, respectively; other measures are averaged within each window. Ensemble means average these run-level summaries across seeds. Comparisons between model versions use differences paired by seed; intervals are pointwise 95% Student-t Monte Carlo confidence intervals across independent seeds. Summaries across regimes give each distinct regime equal weight. Outcome definitions and the procedure for assessing Monte Carlo precision are described in Section IV.A.

Supplementary figures

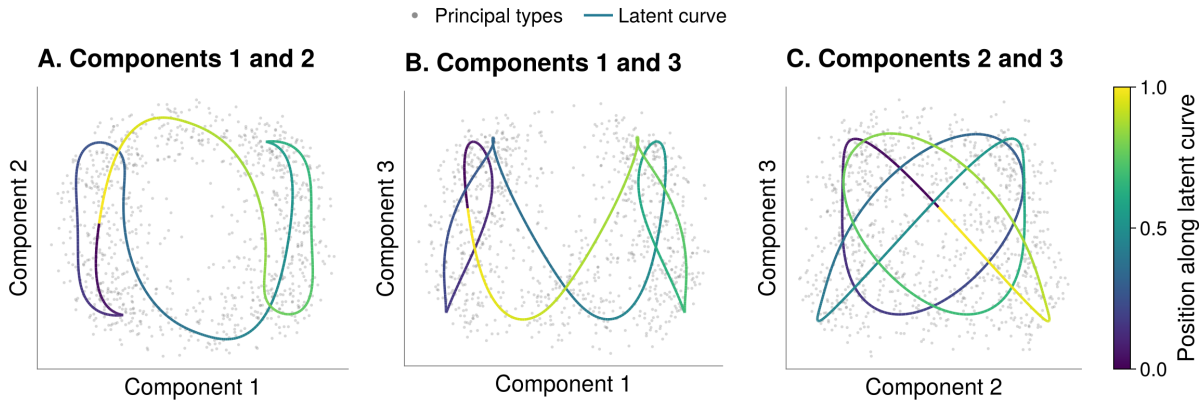


Figure S1: Principal types and their latent curve. All 1000 baseline types from realization 1, projected onto pairs of the curve's first three principal components (A–C). Gray points are realized types; the line shows the latent curve, colored by position along it. Matching and learning use the full eight-dimensional types, not these projections.

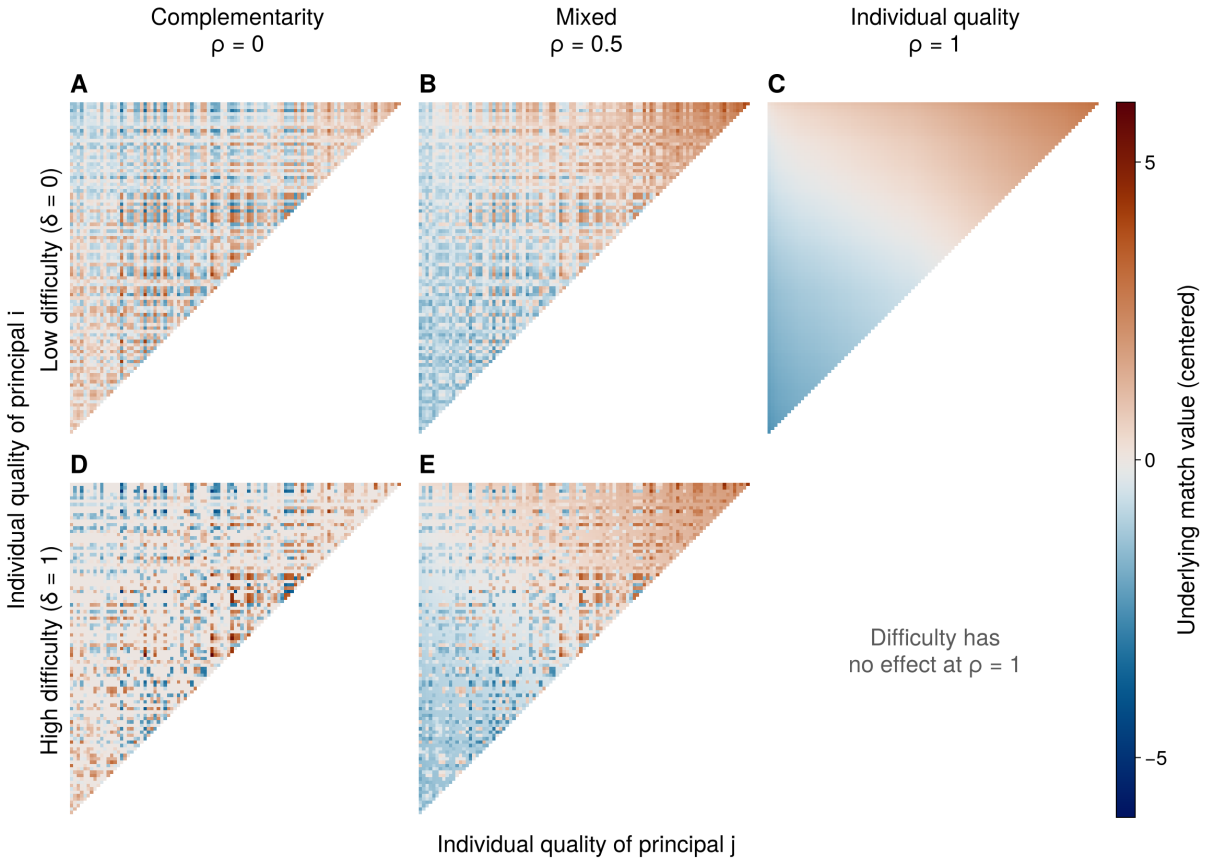


Figure S2: Underlying match value across pairs of principal types. 100 types selected at evenly spaced ranks of individual quality from realization 1 (1000 principals). Types and matching-function objects c, A, and B are fixed across panels. Individual quality increases rightward and upward. Upper triangles exclude self-matches; colors show each pair's underlying value minus the grand mean of the full value matrix.

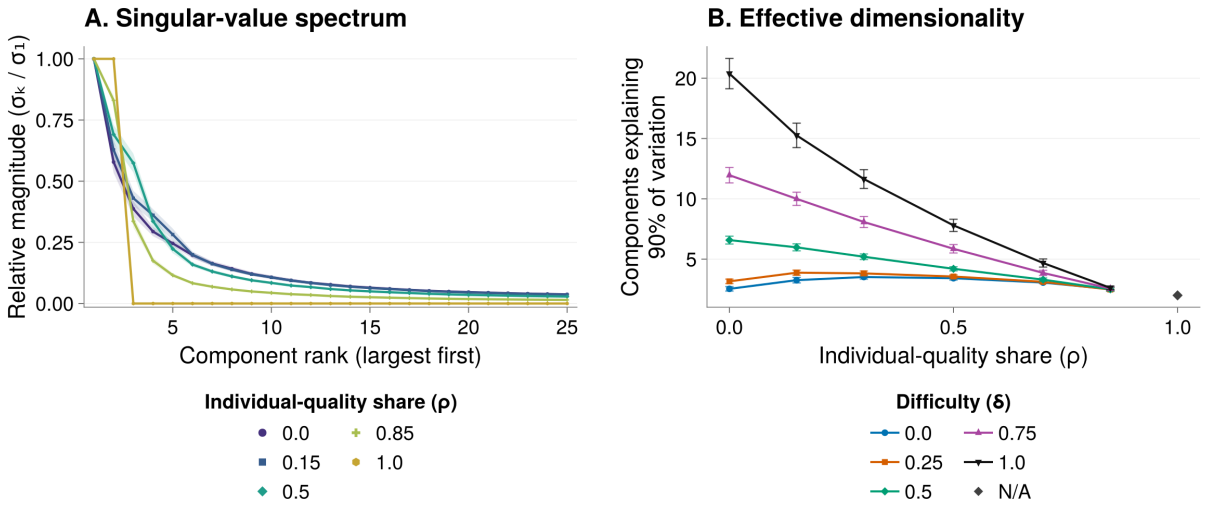


Figure S3: Dimensionality of underlying match value. Matrices of pair-specific underlying match values, centered within each realization and summarized over 50 independent realizations. Types and matching-function draws are fixed within each realization as ρ and δ vary. A: the first 25 singular values at $\delta = 0.5$, divided by the largest within each realization. B: the fewest components accounting for 90% of the sum of squared singular values. Lines and markers are means; ribbons and whiskers show pointwise 95% Monte Carlo intervals. N/A denotes $\rho = 1$, where underlying match value depends only on individual quality and difficulty is inapplicable.

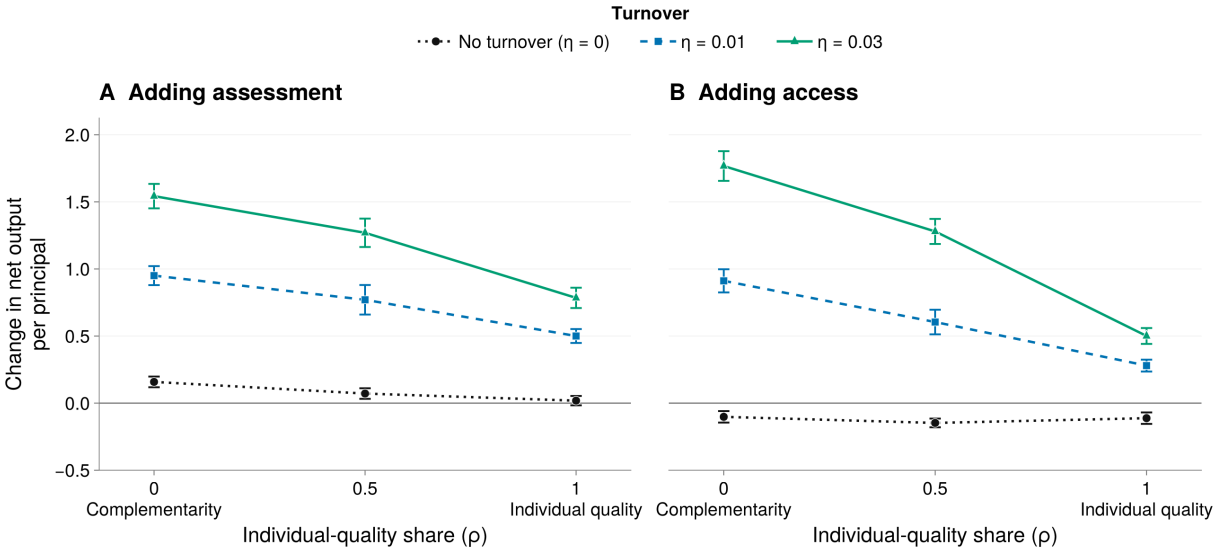


Figure S4: Contributions of assessment and access across matching composition. Late-mean differences paired by seed: A, full service minus access-only; B, full service minus assessment-only. Parameters other than matching composition and turnover remain at baseline. Points average 20 seeds per regime; bars show pointwise 95% Monte Carlo intervals. Connecting lines are visual guides.

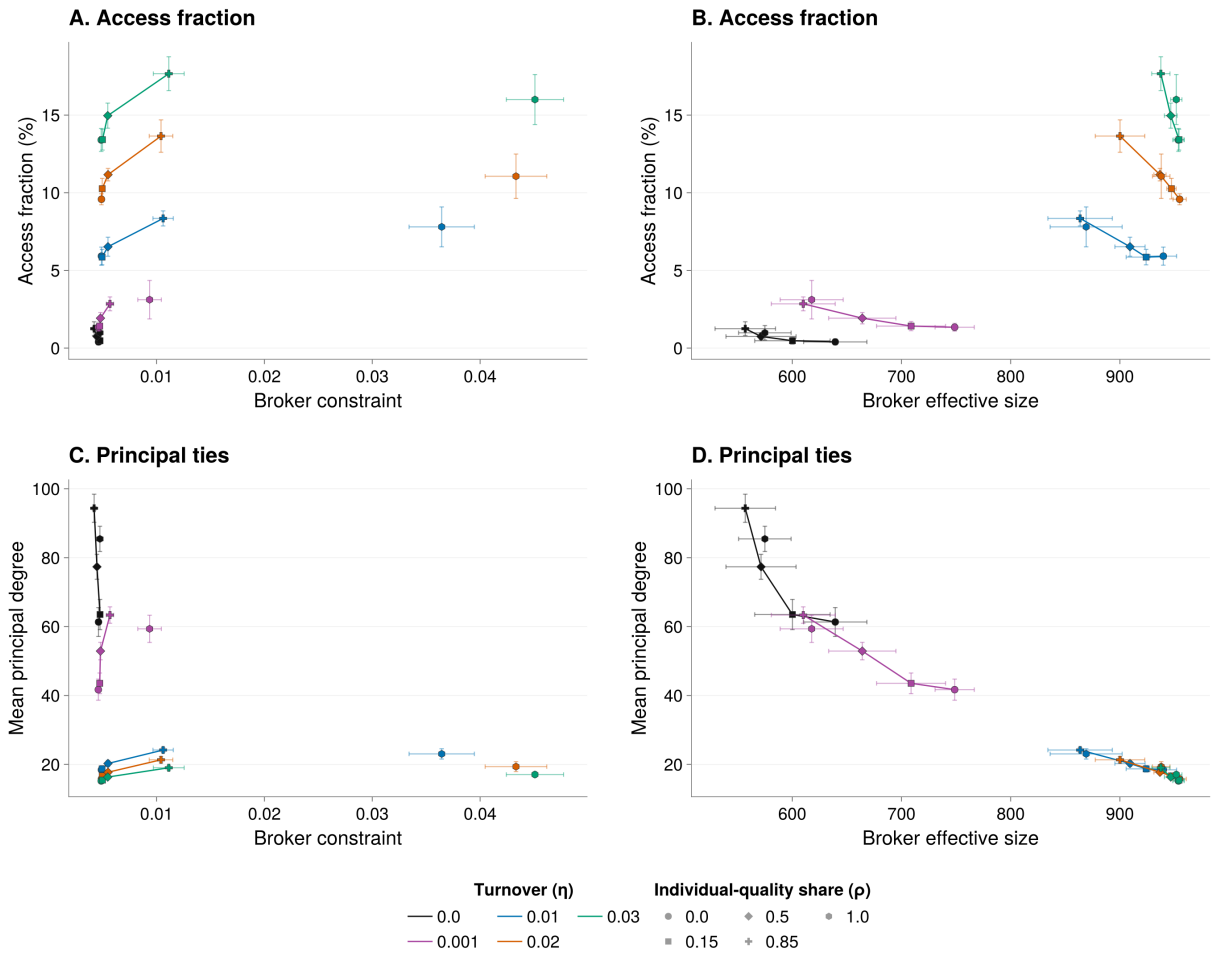


Figure S5: Structural position and access under alternative measures. Ensemble means from the neural-network simulations, using constraint and effective size at period 500 and late-window averages for access and degree. Parameters other than matching composition and turnover remain at baseline. Lines connect matching compositions at fixed turnover, with $\rho = 1$ shown separately. Principal degree includes the broker tie when present. Whiskers show pointwise 95% Monte Carlo intervals.

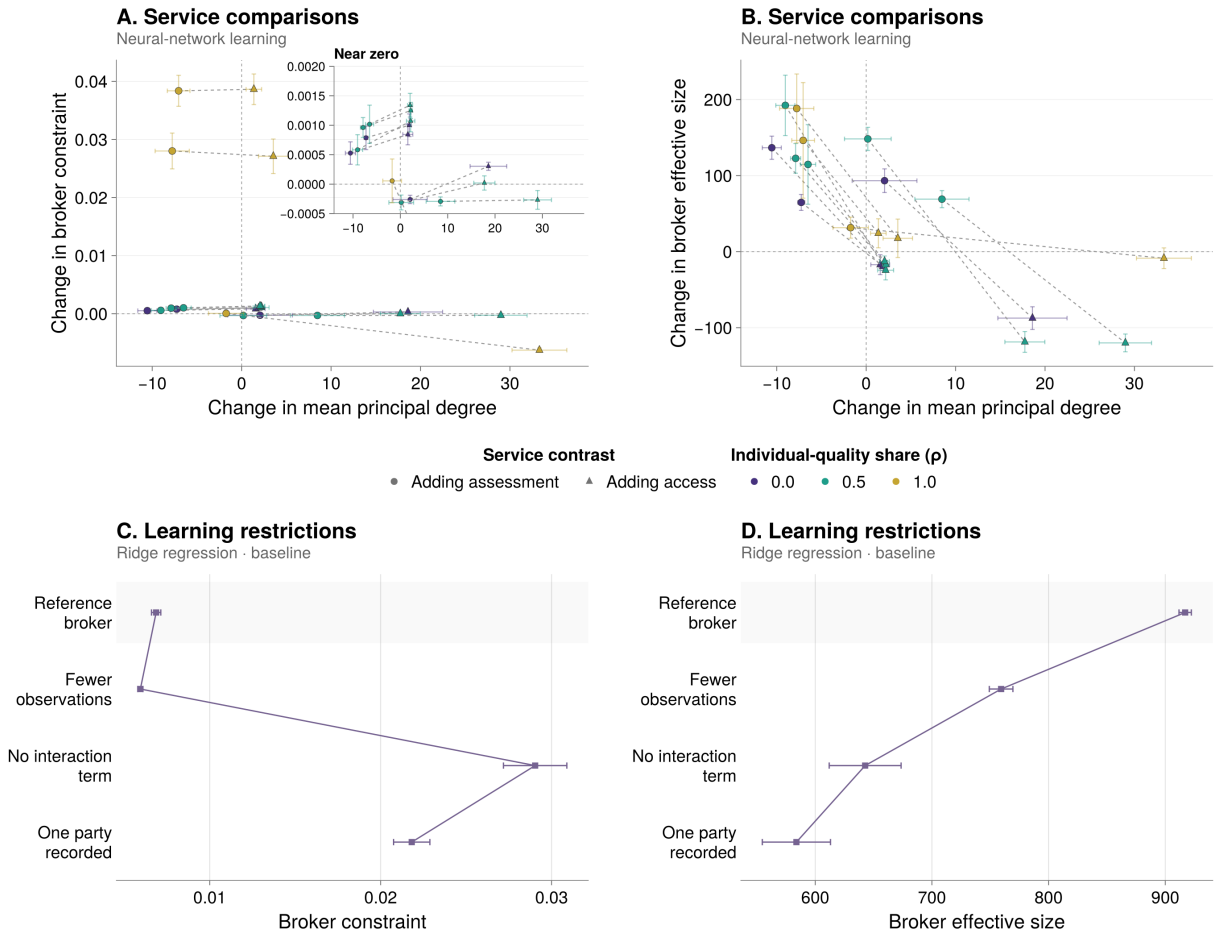


Figure S6: Structural consequences of services and learning under alternative measures. A–B: differences paired by seed across matching composition and turnover, with other parameters at baseline. Adding assessment is full service minus access-only; adding access is full service minus assessment-only. Dashed lines connect contrasts within each regime; the inset enlarges the region near zero. C–D: ensemble means from the baseline ridge experiments. Constraint and effective size use period 500; other measures use late-window averages. Whiskers show pointwise 95% Monte Carlo intervals.

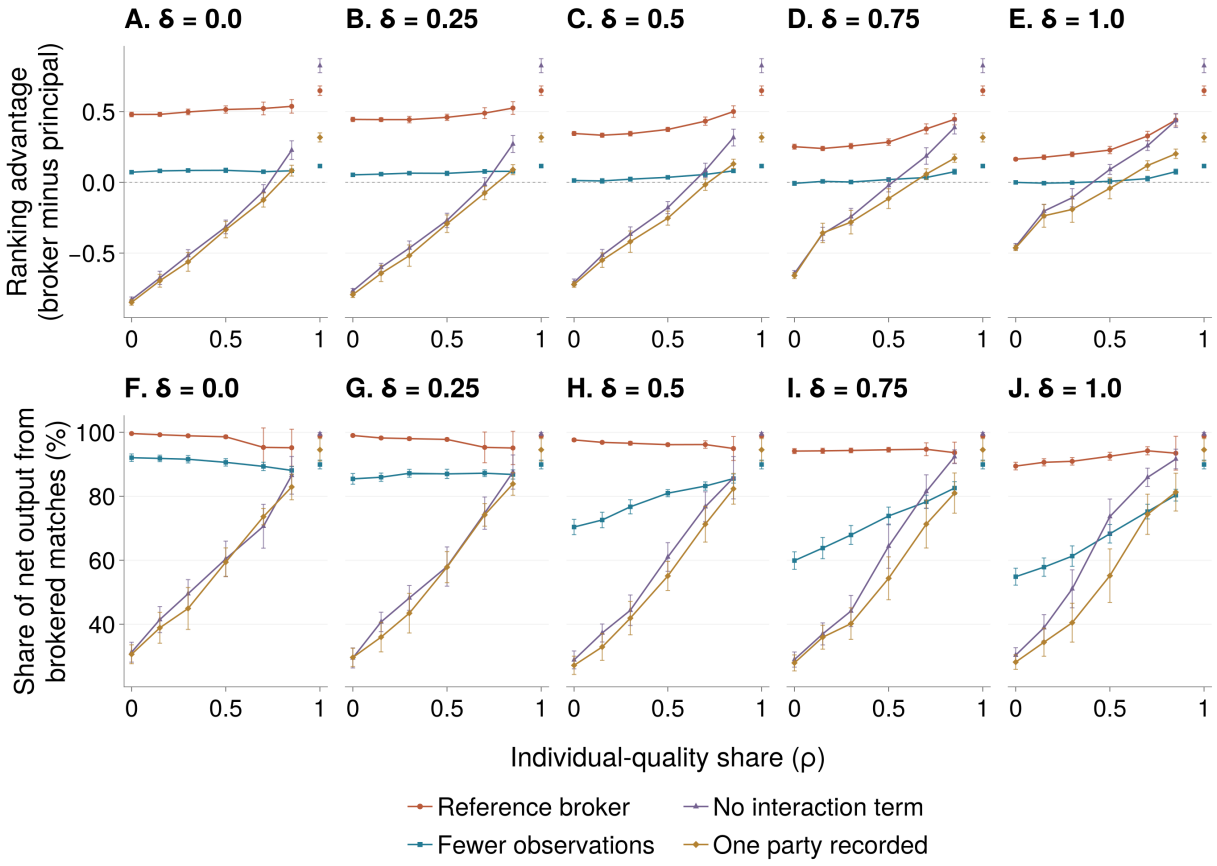


Figure S7: Sources of the broker's advantage across matching difficulties. Ensemble late means from the ridge experiments; parameters other than ρ and δ remain at baseline. Ranking differences and net-output shares are calculated within seed before averaging. Whiskers show pointwise 95% Monte Carlo intervals. The same $\rho = 1$ estimates appear in each column because difficulty has no effect there. Lines are visual guides.